

The role of transport systems in housing insecurity

A mobility-based analysis

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Housing insecurity

A household is **housing insecure** when it cannot reliably access or remain in housing that is **affordable, stable, safe and adequate**.



Affordability

Housing costs absorb too much income



Stability

Eviction risk, forced moves or uncertainty



Quality & safety

Poor facilities, overcrowding or unsafe conditions

Housing policy is not the whole mechanism

A household may find cheaper or safer housing, but still face a second barrier: the transport system may not connect vulnerable residential areas to employment areas that support upward mobility.





Research question 1

How can sociological definitions of housing insecurity be reconciled with empirical approximations to provide a comprehensive estimate of housing insecurity?



Research question 2

To what extent does public transportation facilitate job accessibility for households that are vulnerable to housing insecurity?

Datasets

- **Socio-economic data**

- **Housing data:** 2019 American Community Survey tract-level housing variables.
- **Eviction data:** tract-level eviction rates from The Eviction Lab.

- **Transport data**

- **Public transit data:** GTFS feeds from The Mobility Database.
- **Pedestrian network:** OpenStreetMap street/path data processed with UrbanAccess.
- **Driving times:** Openrouteservice estimates using OpenStreetMap road networks.

- **Mobility data**

- **Commuting:** LODES origin-destination commuting flows from the LEHD program.

- Spatial unit: census tracts across 20 US cities.

1. Housing Insecurity Dimensions



Affordability

Captures financial strain using metrics such as severe rent burden, mortgage status, and available housing units per capita.



Safety & Quality

Evaluates structural adequacy based on household access to complete plumbing, kitchen facilities, and telephone service.



Stability

Measures displacement risk through tract-level eviction rates and persons-per-room overcrowding metrics.

1. Housing features

tract-level census + eviction data

Tract ID	Housing feature groups		
	Affordability	Quality & safety	Stability
Tract 1	...	...	...
Tract 2	...	...	...
Tract 3	...	...	...
...	...	...	...
Tract N	...	...	...



Affordability

rent burden, mortgage, housing units



Quality & safety

plumbing, kitchen, phone access

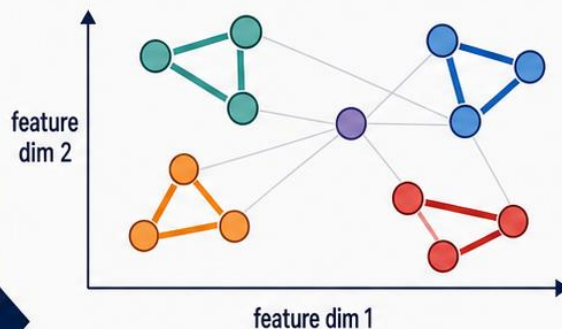


Stability

evictions, overcrowding

2. Build affinity graph

closer feature profiles = stronger edge



Affinity matrix A

	1	2	...	N
1	0	0.8	...	0.1
2	0.8	0	...	0.2
...	...	...	...	...
N	0.1	0.2	...	0

Degree matrix D

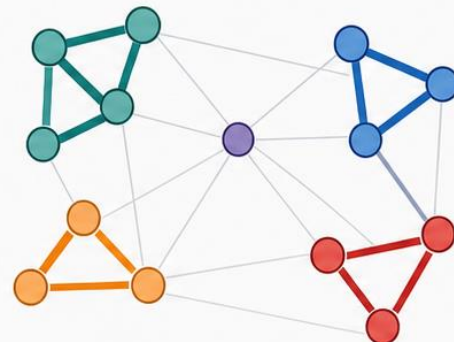
	1	2	...	N
1	2.1	0	0	0
2	0	2.6	0	0
...	0	0	...	0
N	0	0	0	1.7



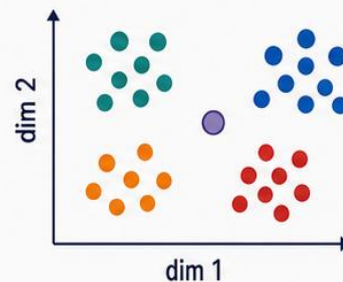
graph Laplacian
from A and D

3. Input to spectral embedding

weighted similarity graph

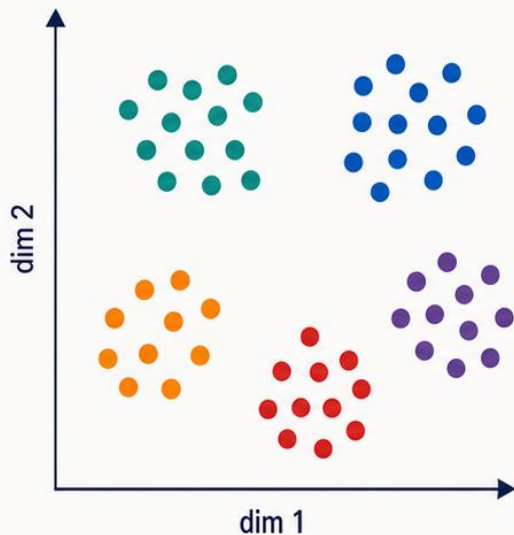


spectral embedding



next:
choose k
and apply
k-means

1. Spectral embedding

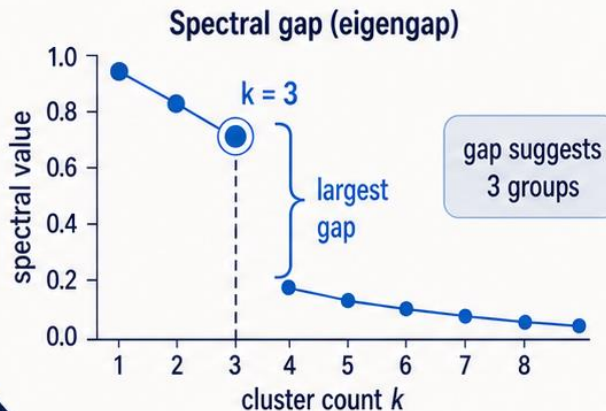


tracts become points



similar housing profiles sit near each other

2. Choose k, then cluster



K-means assignment (k = 3)

assign each tract



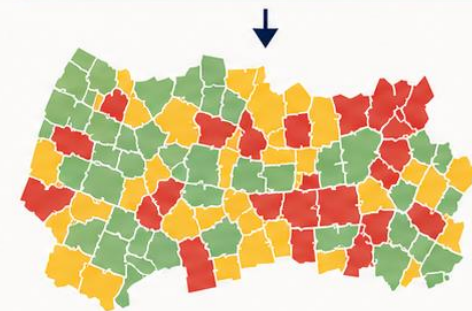
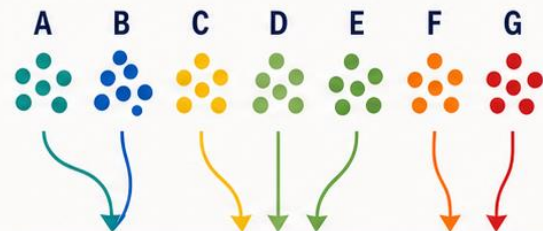
cluster centers



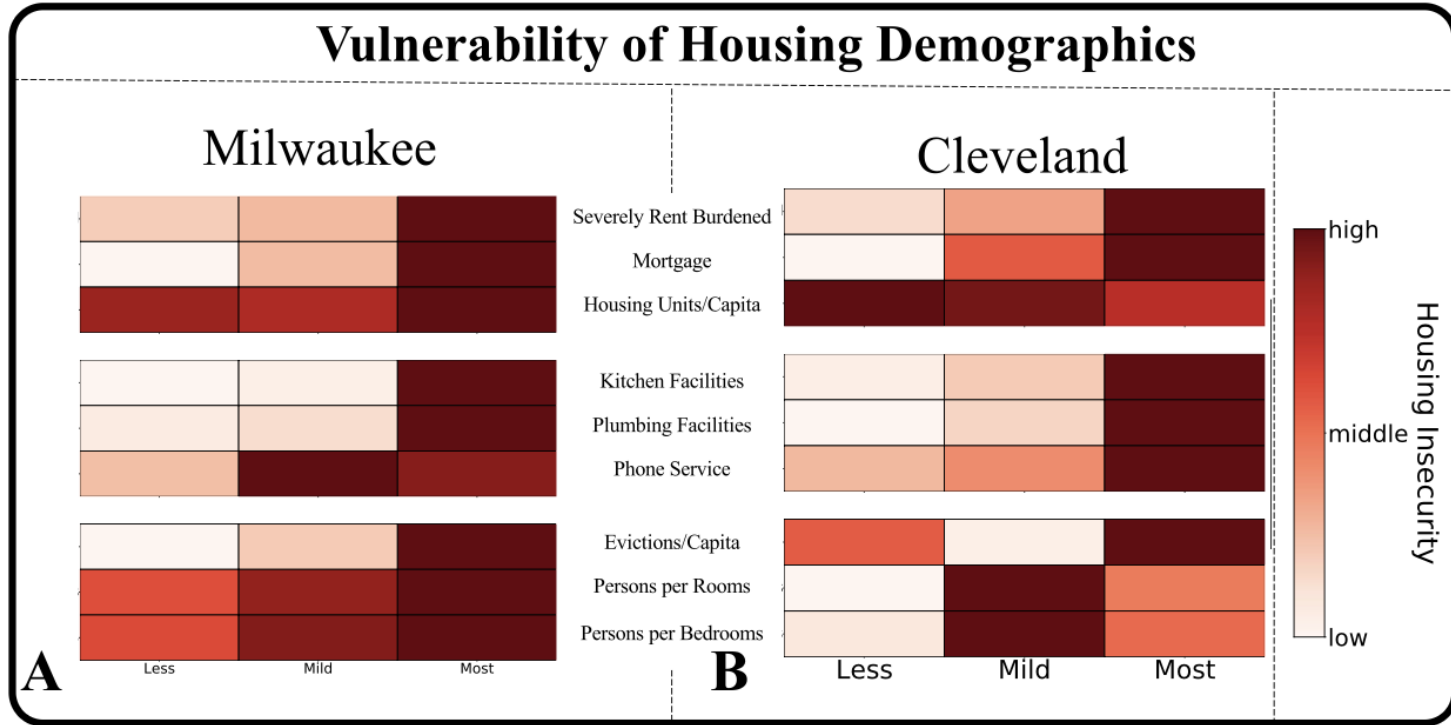
rank cluster averages



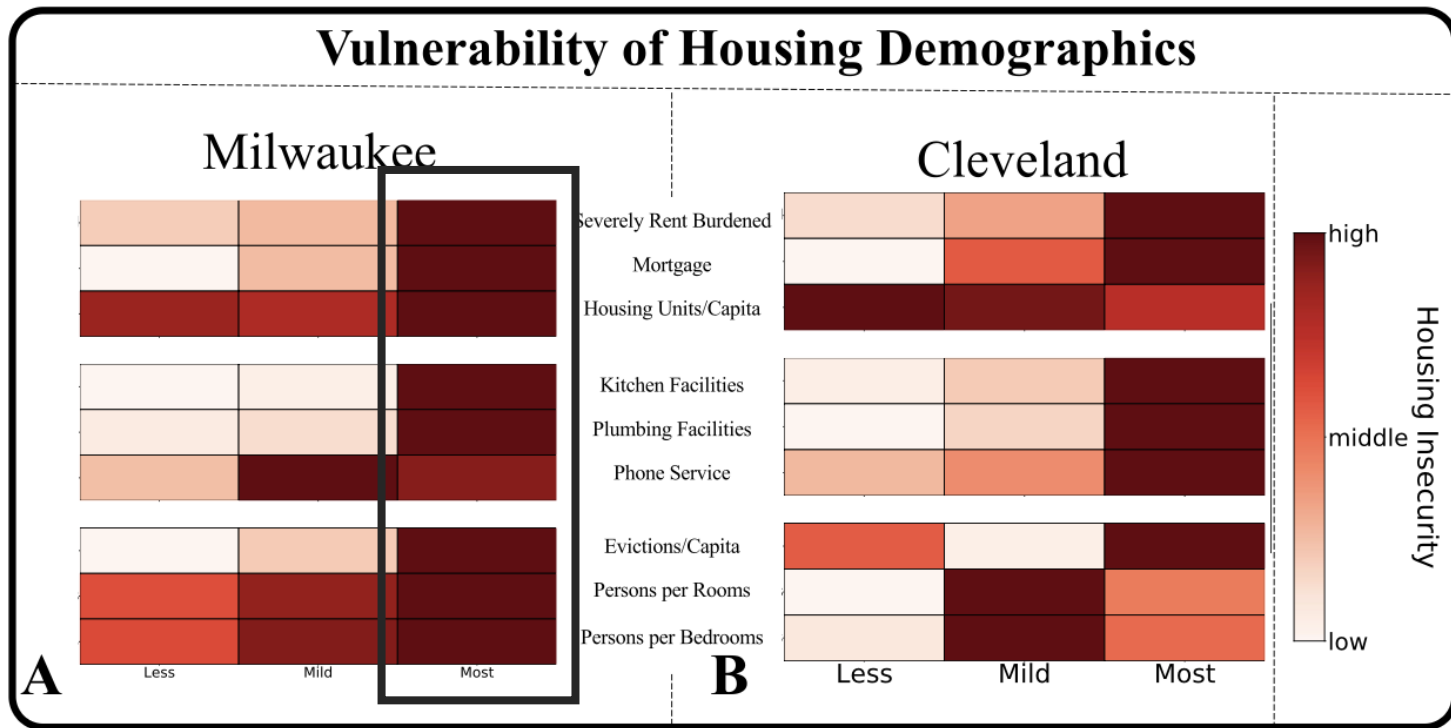
3. Vulnerability levels



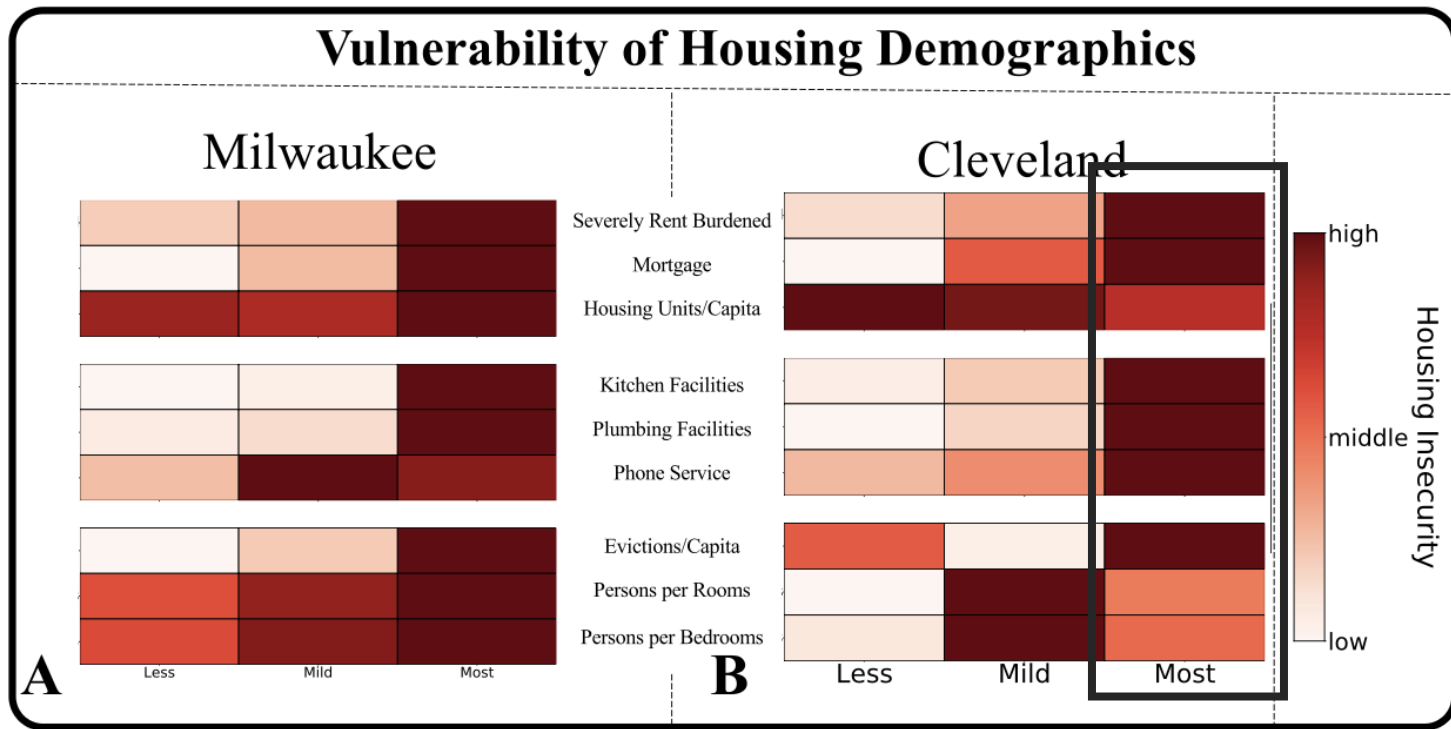
Multidimensional clustering reveals hidden realities



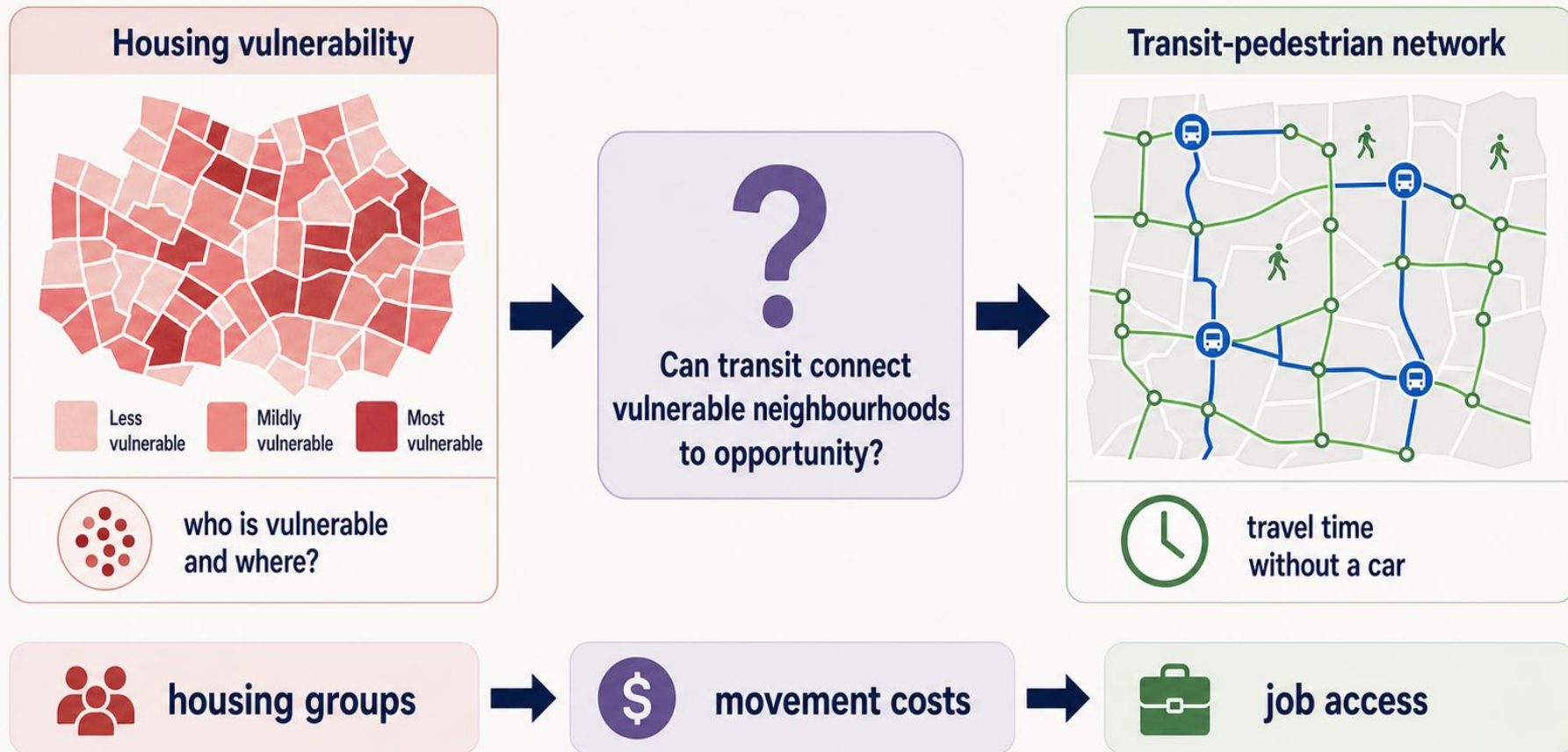
Multidimensional clustering reveals hidden realities



Multidimensional clustering reveals hidden realities



From housing vulnerability to transport burden



Building the transit-pedestrian network

1. Transit schedules

GTFS feeds



calendar.txt



stops.txt

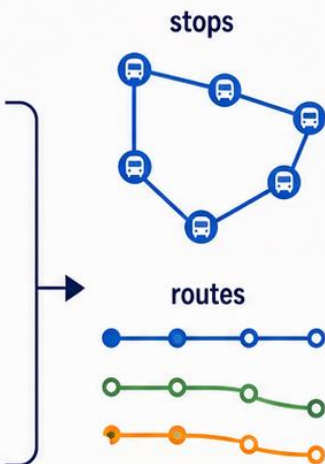


routes.txt



stop_times.txt

...

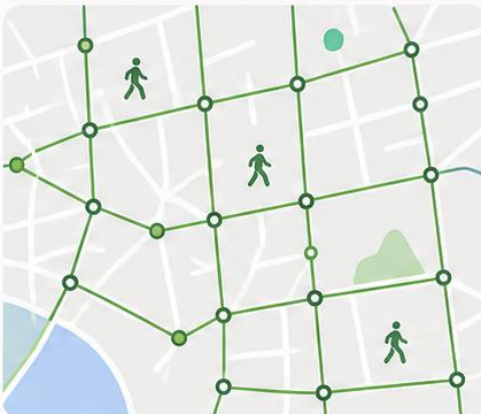


scheduled travel times

stop	trip	arrival	departure
A	T1	08:15	08:15
B	T1	08:18	08:19
...	...	...	...
C	T1	08:27	08:27

2. Pedestrian network

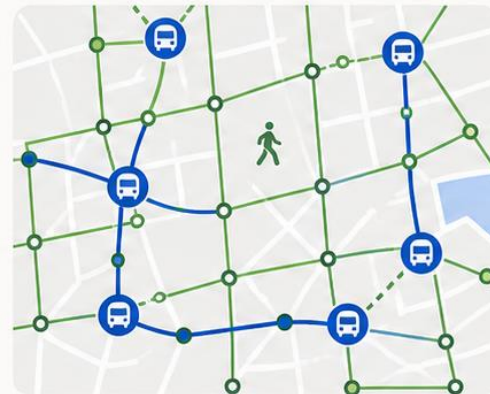
OpenStreetMap



- pedestrian nodes
- walking links
- 3 mph walking speed

3. Combined network

UrbanAccess joins the layers



- transit nodes (stops)
- nearest pedestrian node
- transit edges (scheduled travel times)
- walking edges (3 mph)

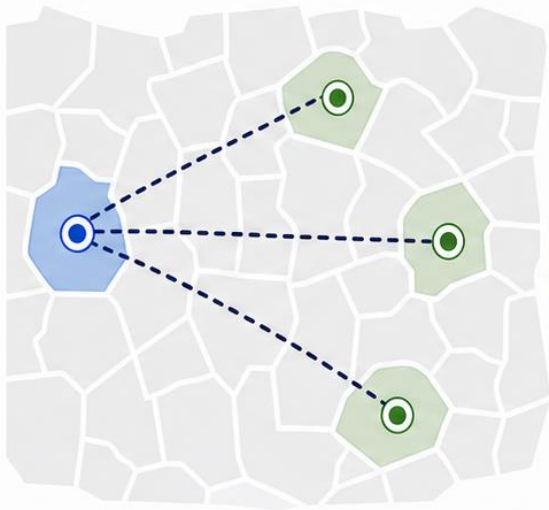


tract-to-tract transit time

Measuring transit efficiency

1. Tract-to-tract trips

all origin-destination tract pairs

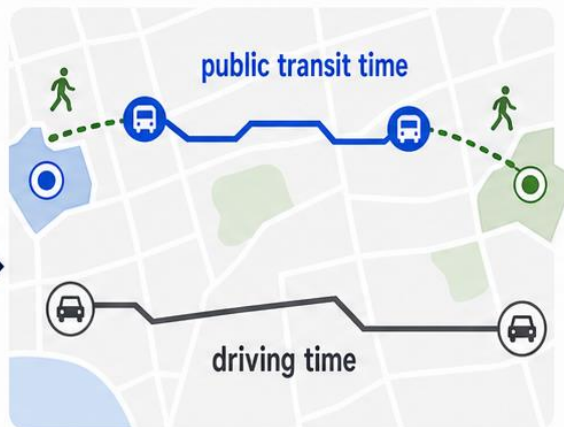


Monday rush hour

06:30-10:30

2. Compare travel times

same origin and destination tract



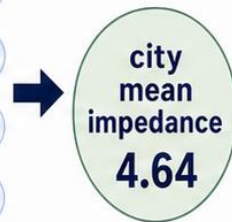
$$\text{travel impedance} = \frac{\text{transit time}}{\text{driving time}}$$



driving time is the baseline
no congestion

3. City-level efficiency

average over all tract pairs



lower ratio =
more efficient

higher ratio =
less efficient



Milwaukee 1.90



Median city 4.95



Fort Worth 7.26



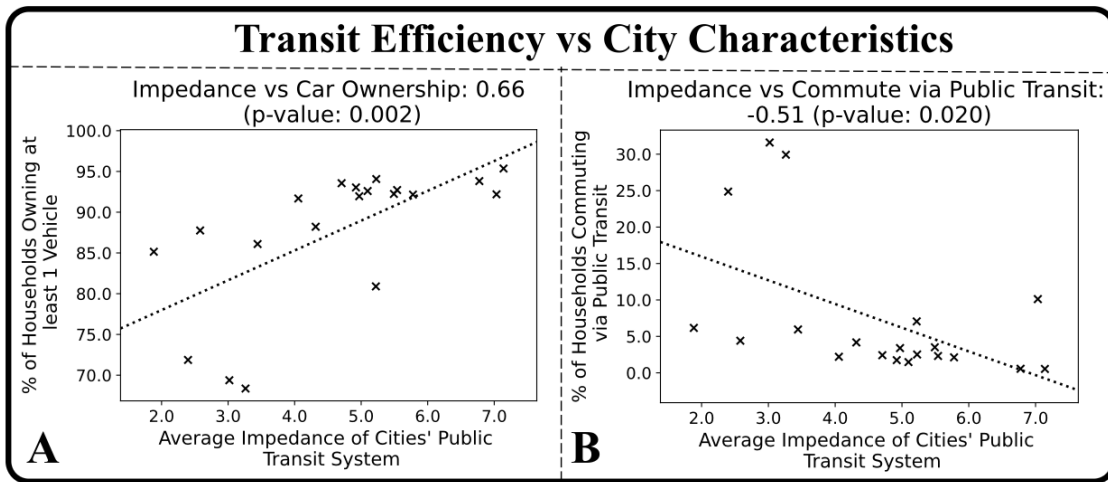
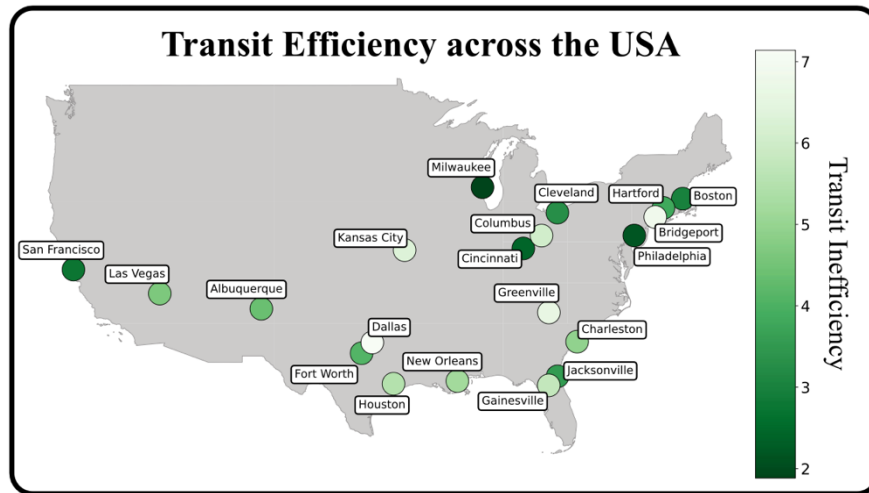
more efficient

less efficient

From impedance to transit efficiency

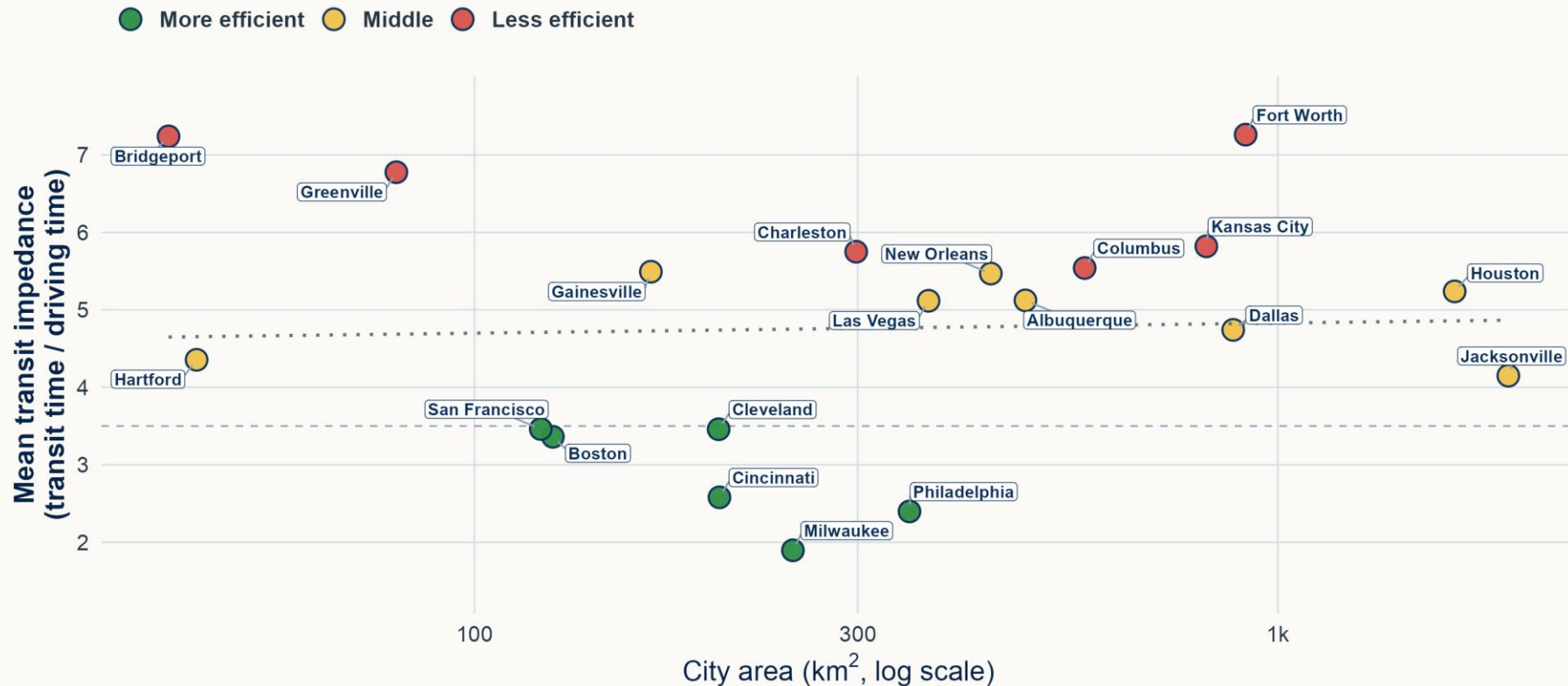
- Average impedance over all pairs of census tracts

$$\text{eff}_c = \frac{\sum_{t_1, t_2} \mathcal{Z}_{t_1, t_2}}{|T|^2}$$



More efficient transit is not simply a function of city size

Compact cities can be efficient, but larger cities can be too; similar-sized cities vary widely.



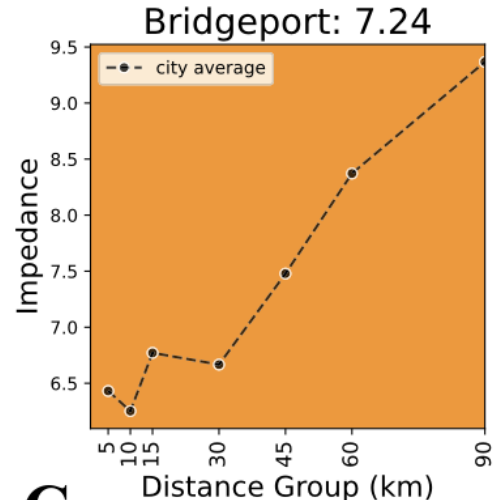
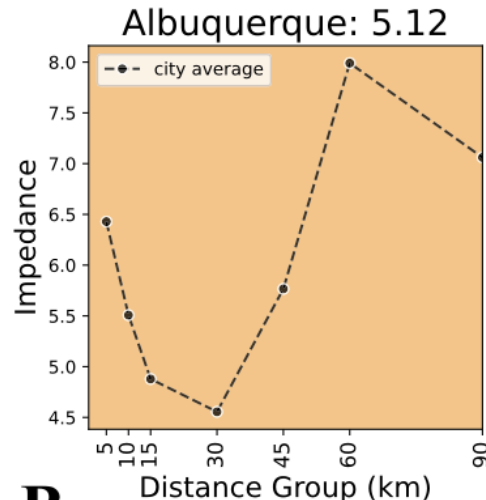
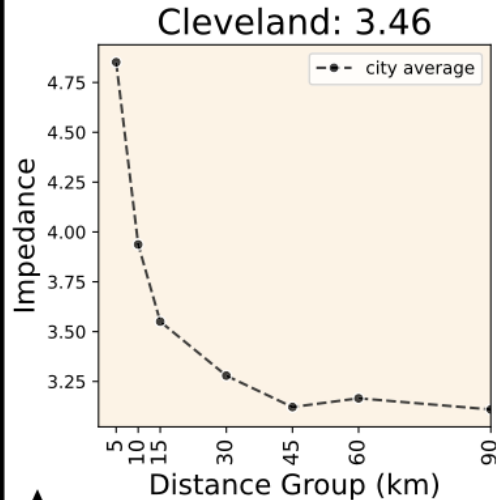
Lower impedance = more efficient public transit relative to driving.

Efficiency vs commuting distance



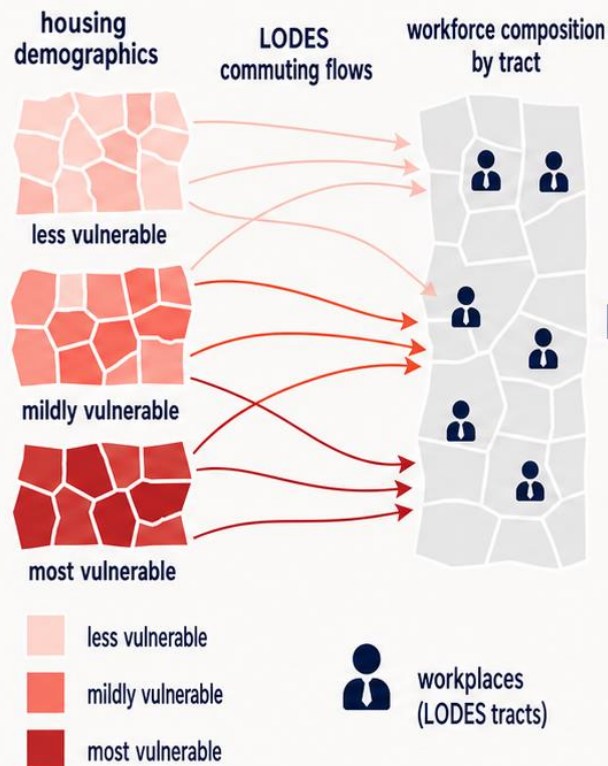
University
of Exeter

Transit Efficiency as a Function of Distance

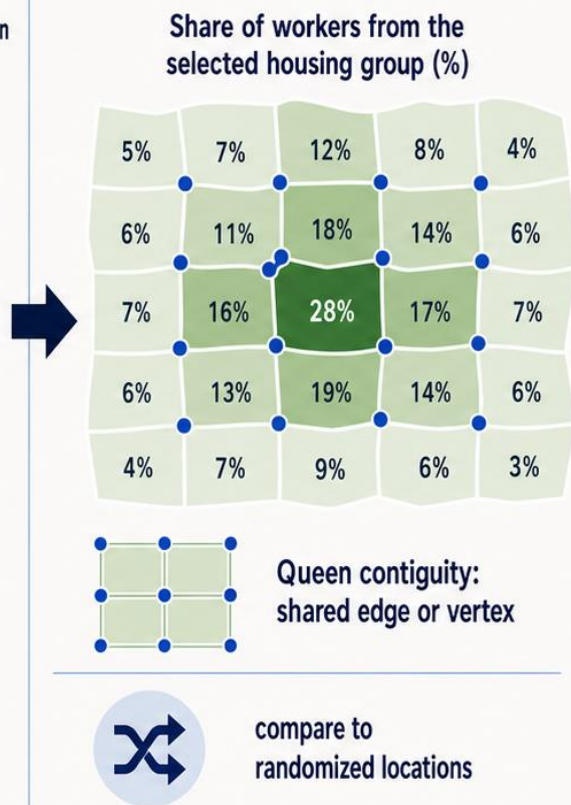


Defining employment landscapes

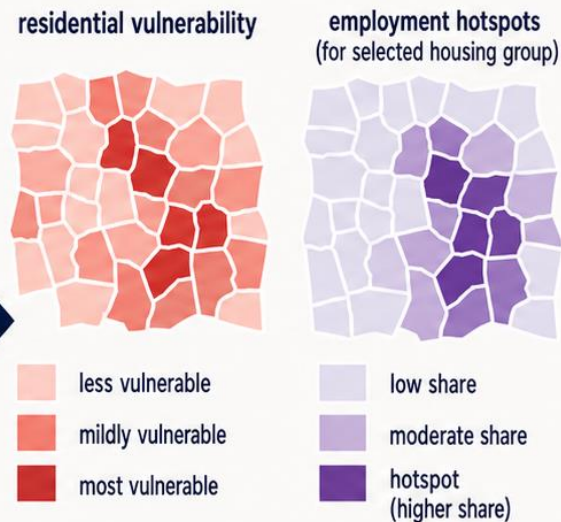
1. Link homes to workplaces



2. Test spatial clustering



3. Identify employment hotspots



LISA: local clusters
high employment share +
high nearby share

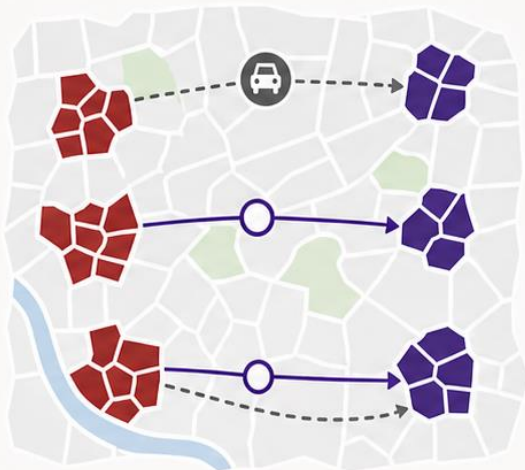


used later to connect
housing, jobs and transit

Simulating potential avenues for social mobility

1. Empirical commutes

observed home-work links

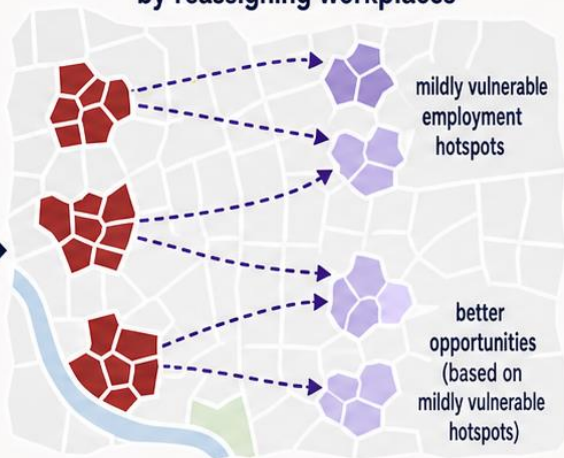


 most vulnerable home tracts  most vulnerable employment hotspots

 empirical driving time  empirical transit time

2. Reassign workplaces

simulate new links to better opportunities by reassigning workplaces



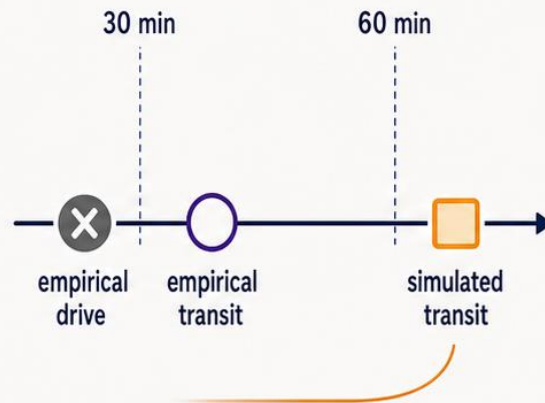
 most vulnerable home tracts  mildly vulnerable employment hotspots  better opportunities (based on mildly vulnerable hotspots)

 randomly sample hotspots
1000 iterations

This is a simulation, not observed data.

3. Compare commute times

median commute times (example)

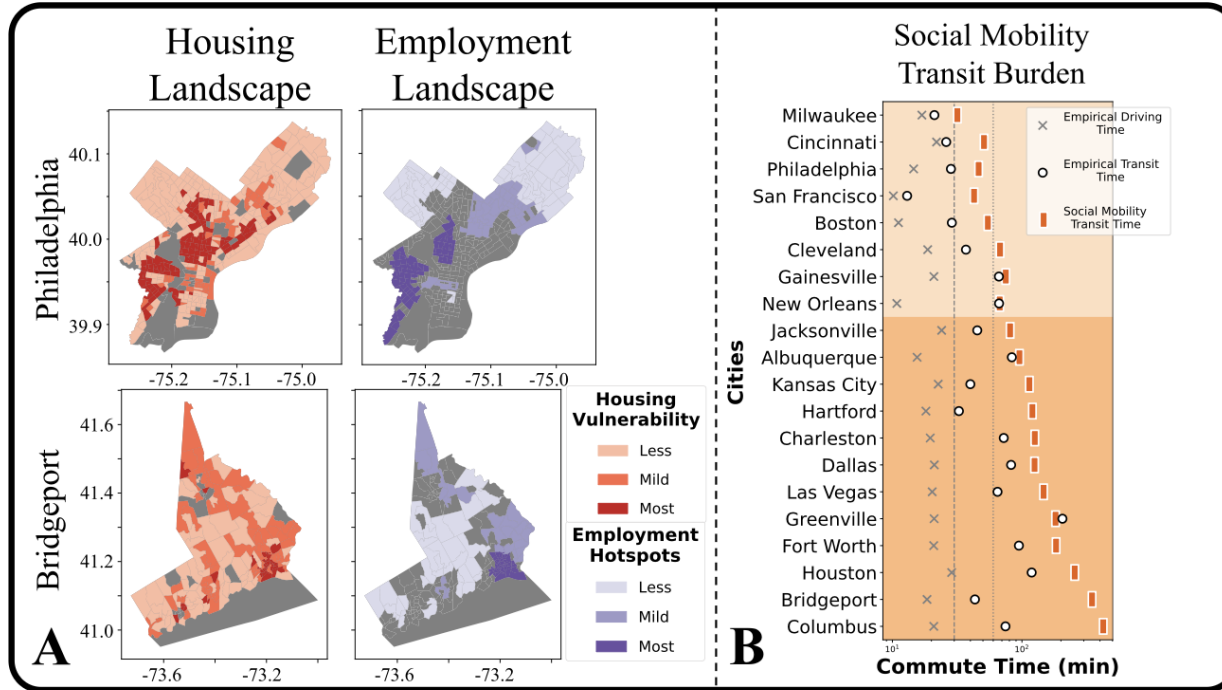


in many cities, simulated transit commutes are much longer and often exceed 60 minutes



better job access can mean much longer transit commutes

Transit can turn opportunity into a long commute

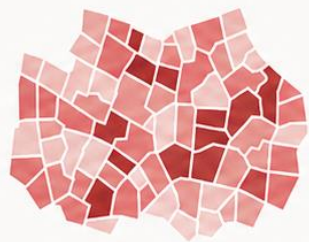


Key takeaway



Transport can reinforce housing insecurity when it limits access to better job opportunities.

1. Housing insecurity is multidimensional



Less vulnerable Mildly vulnerable Most vulnerable



Affordability



Quality / Safety



Stability

vulnerability is measured across several housing conditions

2. Transit burden is uneven



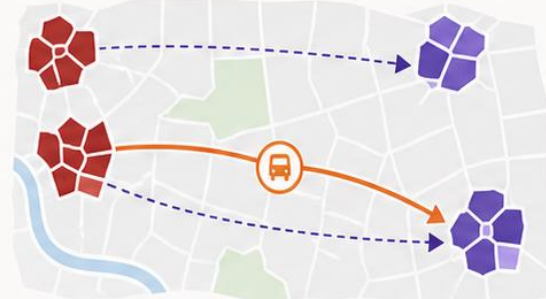
75 min



25 min

public transit often takes much longer than driving

3. Opportunity has a geography



better job access can imply longer transit commutes



Housing, jobs and transport must be analysed together.

Acknowledgement

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